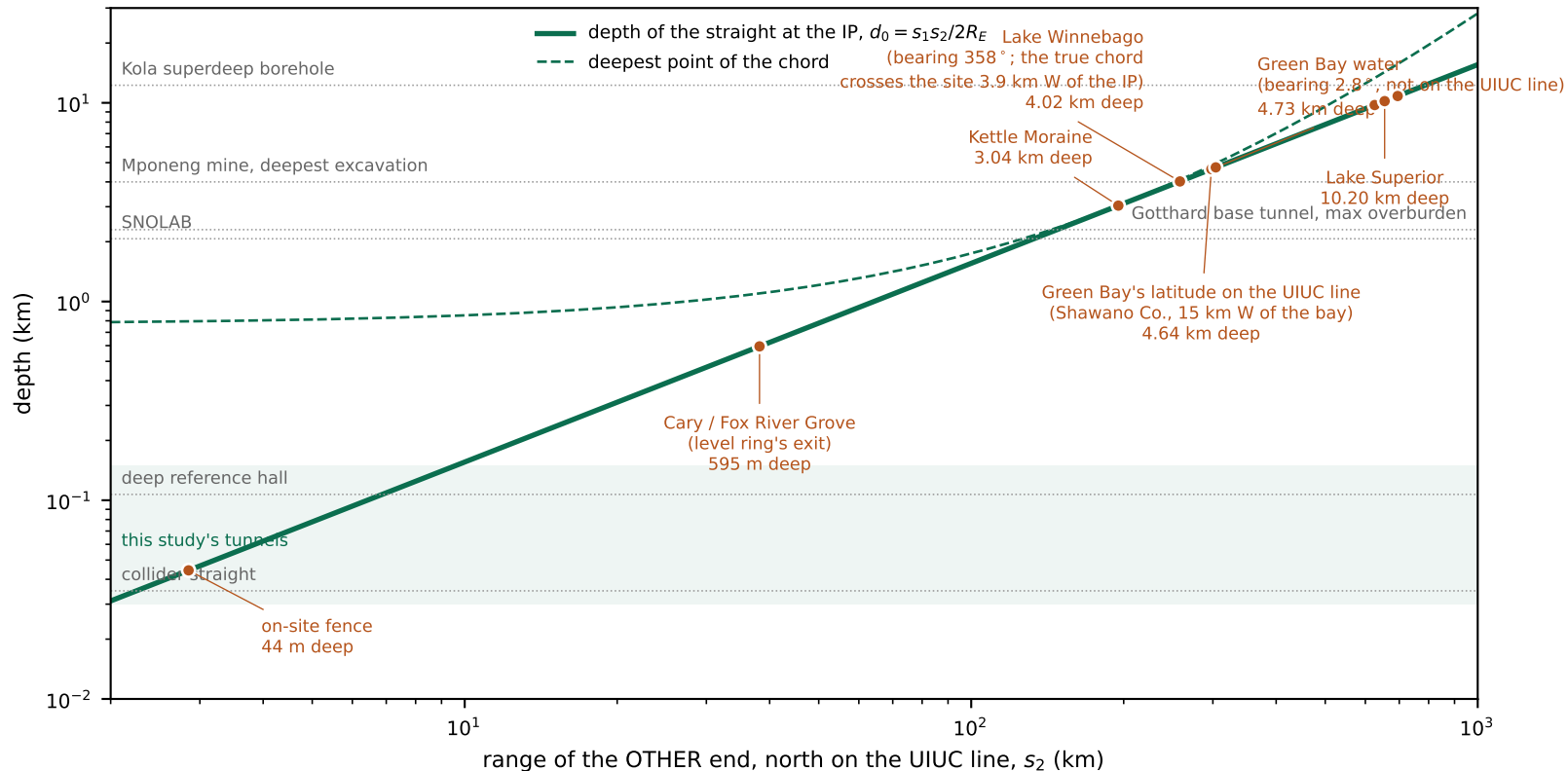


One straight with BOTH ends far: south end at UIUC (198 km), north end at s_2
the straight must sit $d_0 = s_1 s_2 / 2R_E$ below the IP – kilometres, for any northern water



A straight is a chord: its two exits obey $s_{near} s_{far} = 2R_E d_0$. Pushing the near exit from 2 km to 300 km (Green Bay's latitude) takes the straight from 35 m to 4.7 km;

Lake Superior needs 10 km, deeper than any mine. The feasible route is the ring's two straights aimed separately at ordinary depth (see text): inclined arcs and ~ 142 mrad of vertical bending per lap.